

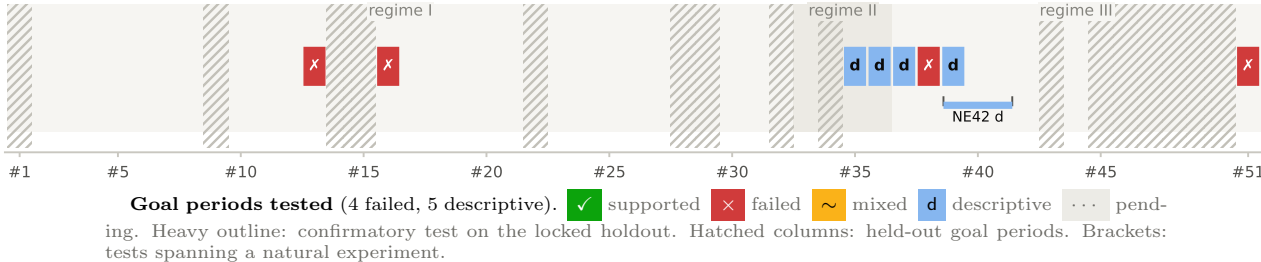
H142: The content response to aligned reads saturates as a Langevin-function torque

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-07

Complete 50% Faithfulness 1/5 Usefulness 1/5

Bottom line. The step toward a direction does not grow with the number of aligned reads at fixed batch size and topic; the Langevin form does not beat a line out of fold, and only the newest item pulls.

level: hypothesis · pipeline checks 71% · updated 2026-10-07



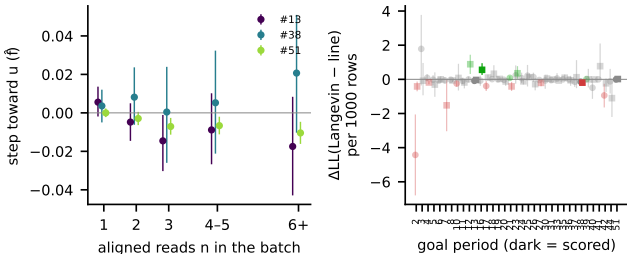
WHAT HAPPENED

H142 asks whether the content step toward a direction saturates with the number of aligned reads in one batch, as a classical moment saturates in a field (the Langevin function). Rows are (talk call, direction) pairs; directions are day-fold clusters of reader statements. Synthetic checks on real skeletons showed that the card’s form mistakes H113’s linear channel for saturation (60–100% of runs), so Amendment A1 lets the amplitude vary with batch size. On the powered, field-identified periods (bge #13, #38, #51; gte adds #16) the step does not grow with aligned reads at all: the pooled out-of-fold $\Delta LL(\text{Langevin} - \text{line})$ is $-9.0 [-17.2, -2.2]$ nats (bge), and $\hat{f}(1) = 0.000 [-0.002, 0.002]$ on #51. The pre-set kill fires. Only the newest batch item pulls.

PHYSICAL MODEL

Model 11 (vector spins, mean-field equation of state). The reader’s projected statement is a moment; each aligned read adds to a field along a direction u , one of $K = 8$ day-fold cluster centroids. n : aligned reads (cosine ≥ 0.3) in the ledger batch of size k . α_c : call effect; δ : direction $\times$ room $\times$ hour effect (hot topic). Rivals: a line in n (H113 at fixed k), a power law, the newest item.

$$y_{c,u} = \alpha_c + \delta_{u,\text{room},h} + a_{b(k)} h(n_{c,u}) + g(n_{c,u}^F) + \beta_{\text{new}} \text{newest}_{c,u} + \beta_{\text{name}} n_{c,u}^{\text{name}} + \epsilon$$
$$h(n) \in \{n, n^p, L(cn)\}, \quad L(x) = \coth x - 1/x, \quad n_{\text{sat}} = 3/c, \quad \Delta_{\text{curv}} = \ln \frac{f(2)}{f(1)} - \ln \frac{f(6+)}{f(3)}$$



Observables. Left: the model-free step curve $\hat{f}(n)$ (95% agent-day cluster bootstrap) on the three scored bge periods; it is flat at zero, not a saturating rise. Right: out-of-fold $\Delta LL(\text{Langevin} - \text{line})$ per 1000 rows for every period (circles bge, squares gte; dark = scored; green CI > 0 , red < 0 , gray includes 0).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator the answer is simple. Repeating the same point in one batch does not move an agent further toward it: the second to sixth aligned message adds nothing measurable once the batch size and the topic of the hour are held fixed. What moves the next statement is the newest message ($+0.010 [0.007, 0.014]$ per direction on #51). Put the key point last; do not send it several times in one batch. The Langevin picture is not needed to say this; the curve has no rise to saturate.

RESULTS VS PREDICTIONS

Prediction	Observed (bge; gte)	Verdict
P1 pooled $\Delta LL(L - \text{line}) > 0$	$-9.0 [-17.2, -2.2]; +6.1 [-4.1, 15.7]$	failed (kill)
P2 $\Delta_{\text{curv}} > 0$	undefined: no positive $\hat{f}$ bins	failed
P3 $n_{\text{sat}} \in [1.5, 6]$	37 [3.4, 407]; 12 [1.8, 77]	failed
P4 read beats in flight	$-0.0025 [-0.0078, 0.0029]$	failed
N1 #51 wakes	$-5.3 [-13.3, 1.0]; -7.4 [-18.7, 0.1]$	failed
N2 NE42 n_{sat} fixed	n_{sat} unidentified, sides unpowered	descriptive
S1 power at #51	1.00, size 0.02 (after A1)	supported
S2 Δ_{curv} size	≤ 0.01	supported
S3 $< 1/2$ powered	3/6; 4/7	failed

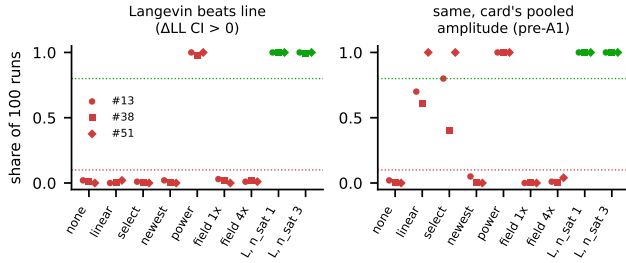
GOAL PERIODS IN DETAIL

G13	× failed
G16	× failed
G35	d descriptive
G36	d descriptive
G37	d descriptive
G38	× failed
G39	d descriptive
G51	× failed
NE42	d descriptive

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic worlds on the real skeletons of #13, #38, #51 (100 runs each). Left: with batch-size amplitudes (A1), the Langevin worlds pass (power 0.99–1.00) and the linear, selection, newest-item and field worlds stay ≤ 0.03 . The power-law world also beats the line; $\Delta LL(L - \text{power})$ separates it (0.00). Right: the card's original single amplitude passes the linear and selection worlds in 40–100% of runs.

CAVEATS

Directions are clusters of reader statements, a coarse ruler; an aligned read is one with cosine ≥ 0.3 to a centroid. The direction $\times$ room $\times$ hour effects and the window-field projection could absorb real uptake that follows the hour's topic; post hoc variants without them keep the flat curve. The result is a null at the H113-calibrated effect size (power ≥ 0.87), not a bound on tiny effects. #16 (gte only) shows a positive ΔLL with no positive bin. Reserved periods were not run.

ROUND 2

Decompose H113's batch uptake into newest, aligned-cluster and off-cluster parts on the same calls; fit the newest-item pull against age and rank.

- **H142-R1** Newest-item channel.
- **H142-R2** Named aligned items.
- **H142-R3** Where does H113's uptake go?

H142-langevin-torque-saturation · generated 2026-10-07 10:14 by infra/summaries/build_summaries.py · card hypotheses/H142-langevin-torque-saturation/README.md · ratings: H142 round-1 agent (suggested) (page 2)